

Med-R²: Crafting Trustworthy LLM Physicians via Retrieval and Reasoning of Evidence-Based Medicine

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Abstract

Large Language Models (LLMs) have exhibited remarkable capabilities in clinical scenarios. Despite their potential, existing works face challenges when applying LLMs to medical settings. Strategies relying on training with medical datasets are highly cost-intensive and may suffer from outdated training data. Leveraging external knowledge bases is a suitable alternative, yet it faces obstacles such as limited retrieval precision and poor effectiveness in answer extraction. These issues collectively prevent LLMs from demonstrating the expected level of proficiency in mastering medical expertise. To address these challenges, we introduce **Med-R²**, a novel LLM physician framework that adheres to the *Evidence-Based Medicine (EBM)* process, efficiently integrating retrieval mechanisms as well as the selection and reasoning processes of evidence, thereby enhancing the problem-solving capabilities of LLMs in healthcare scenarios and fostering a trustworthy LLM physician. Our comprehensive experiments indicate that **Med-R²** achieves an improvement of 13.27% over vanilla RAG methods and even a 4.55% enhancement compared to fine-tuning strategies, without incurring additional training costs. Furthermore, we find that our LLaMA3.1-70B + Med-R² surpasses frontier models, including GPT-4o, Claude3.5-Sonnet and DeepSeek-V3 by 1.05%, 6.14% and 1.91%. Med-R² effectively enhances the capabilities of LLMs in the medical domain.

CCS Concepts

• Information systems → Question answering.

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Keywords

Evidence-Based Medicine, Retrieval Augmented Generation, Large Language Models

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1 Introduction

Large Language Models (LLMs) have emerged as pivotal tools in the medical domain, redefining the contours of healthcare practice [1, 38, 48]. Their ability to process and understand vast amounts of unstructured medical data positions them at the forefront of medical research [7, 32], clinical decision-making [12, 26], and patient care [4, 41]. Despite their remarkable capabilities, LLMs encounter several challenges when applied specifically to healthcare settings:

C1: Inefficiency in Knowledge Acquisition. Existing approaches predominantly rely on training LLMs with datasets from the medical domain, the process of which is inherently compute-intensive and resource-demanding, especially as model sizes increase [31, 37]. Moreover, training with outdated datasets can result in a lack of highly specialized expertise, which can lead to suboptimal clinical recommendations or misinform healthcare professionals [14, 24].

C2: Limited Precision for Medical Retrieval. Compared to domain-specific training, Retrieval-Augmented Generation (RAG) systems provide a cost-efficient solution, leveraging the external knowledge base to enhance content generation [27]. The retrieval quality is critical in RAG systems, where inaccuracy or misinformation can heavily influence the effectiveness of LLMs' augmentation. While efforts [18, 30, 42] have been made to improve the retrieval precision, they neglect the specific and highly professional nature of medical knowledge, where *tailored retrieval enhancement* for distinct medical scenarios remains insufficiently explored.

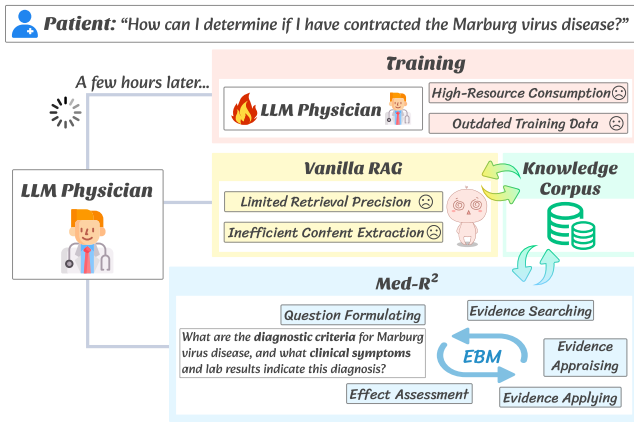


Figure 1: Comparison of Med-R² with existing strategies for medical problem-solving.

C3: Low Effectiveness in Answer Extraction. Considering the constraints imposed by the models' context window length, it is essential to critically appraise the retrieved medical evidence for its validity, impact, as well as applicability, and integrate the most pertinent ones with existing clinical expertise for problem-solving [5, 10]. Nonetheless, current studies fail to develop targeted answer extraction methods tailored for healthcare scenarios, where the nuanced evaluation of evidence hierarchies is required [35].

To address these challenges, we introduce **Med-R²**, a novel medical LLM framework designed in accordance with the principles of Evidence-Based Medicine (EBM), conducting outstanding Retrieval and Reasoning aligned with distinct phases of EBM. **1)** For **C1**, we have established a comprehensive external knowledge base to enhance models' medical performances, offering a more cost-effective and flexible alternative to domain-specific training. **2)** For **C2**, we improve the retrieval precision by refining the original queries according to their respective medical scenarios, while iteratively incorporating chain-of-thought sequences generated from the retrieved content. **3)** For **C3**, we adopt a coarse-to-fine strategy for document appraising and filtering, and select the most pertinent ones supplemented with chain-of-thought demonstrations to assist medical queries addressing. Our contributions are as follows:

- **Challenges in Medical Scenarios.** Through conducting a quantitative analysis of strategies aimed at enhancing models' medical capabilities, we underscore the challenges prevalent in healthcare scenarios, including high computational consumption as well as poor efficiency in knowledge retrieval and extraction.
- **LLM Physician Framework.** We present Med-R², a novel LLM physician framework that integrates the principles of *evidence-based medicine (EBM)* for clinical problem-solving with outstanding retrieval and reasoning capabilities within medical contexts.
- **Performance and Effectiveness.** Extensive experiments indicate that Med-R² achieves a 14.74% improvement over the vanilla RAG methods, and even a 3.32% enhancement compared to the fine-tuning strategies without additional training expenses. Moreover, LLaMA3.1-70B + Med-R² surpasses frontier models for medical problem-solving, achieving average improvements over GPT-4o, Claude3.5-Sonnet and DeepSeek-V3 by 1.22%, 5.33% and 2.80%.

2 Related Work and Discussions

Evidence-Based Medicine (EBM) EBM refers to the application of the best available research to healthcare, which requires evidence integration with clinical expertise and patient values [10, 34, 35]. Clinical questions can be categorized into several types, including *diagnosis, therapy, prognosis, etiology, prevention, cost*, etc. [5]. Each category intersects with EBM principles by emphasizing the collection, evaluation, and application of the best retrieved evidence to inform medical decision-making [11], as detailed in Section A.

LLMs for Medical Domain As the application of LLMs expands, their deployment in the medical domain has become a widely discussed topic [7, 50]. Recent studies have concentrated on the direct use of medical data for the pretraining or fine-tuning of LLMs [36, 38]. Prominent open-source milestones include ChatDoctor [28] which integrates real-world doctor-patient communication data for training, PMC-LLaMA [43] pretrained on 4.9 million medical literature records, and MEDITRON [6], a scaling series of medical pretrained models. However, such extensive training can be computationally intensive. In contrast, Retrieval-Augmented Generation (RAG) systems offer a more efficient alternative, achieving comparable results with reduced training costs and enhancing the model's precision in locating and leveraging knowledge.

Retrieval-Augmented Generation (RAG) The concept of RAG [27] was introduced as a powerful framework for integrating external knowledge into natural language generation tasks, enhancing the accuracy of generated outputs across various domains [9, 17, 51]. In the medical field, RAG has been widely used to improve LLMs' analytical performances by utilizing external knowledge from sources such as medical papers, textbooks, guidelines, and entries [22, 45, 46, 49]. However, while there have been efforts dedicated to optimizing the individual components of RAG pipelines [3, 18, 19, 42], research that integrates the unique requirements of the medical domain remains in its infancy. We incorporate the principles of *Evidence-Based Medicine (EBM)* into medical RAG systems to better address the special demands of healthcare.

3 Med-R²

3.1 Question Formulation

In the medical domain, the efficacy of information retrieval is closely tied to the professionalism of the query. A well-crafted, professional query that includes precise descriptions of medical symptoms can markedly enhance the accuracy and relevance of the documents retrieved from knowledge bases. Conversely, non-standard terms and isolated numerical values frequently hinder effective information retrieval. Moreover, the focus of the desired response varies depending on the type of clinical consultation. For instance, in queries pertaining to *etiology*, users seek insights into potential causes of a condition, including risk factors, pathogens, or genetic predispositions. In contrast, for *prognosis*-related queries, users are interested in understanding the long-term outcomes or patient prognoses, such as survival rates or recurrence probabilities.

The clinical question formulation stage consists of two components: **query classifier** and **query reformulator**. The query classification encompasses two dimensions: *Evidence-Based Medicine (EBM) categories* and *general natural language question types*. Specifically, we have delineated six distinct EBM categories and twelve

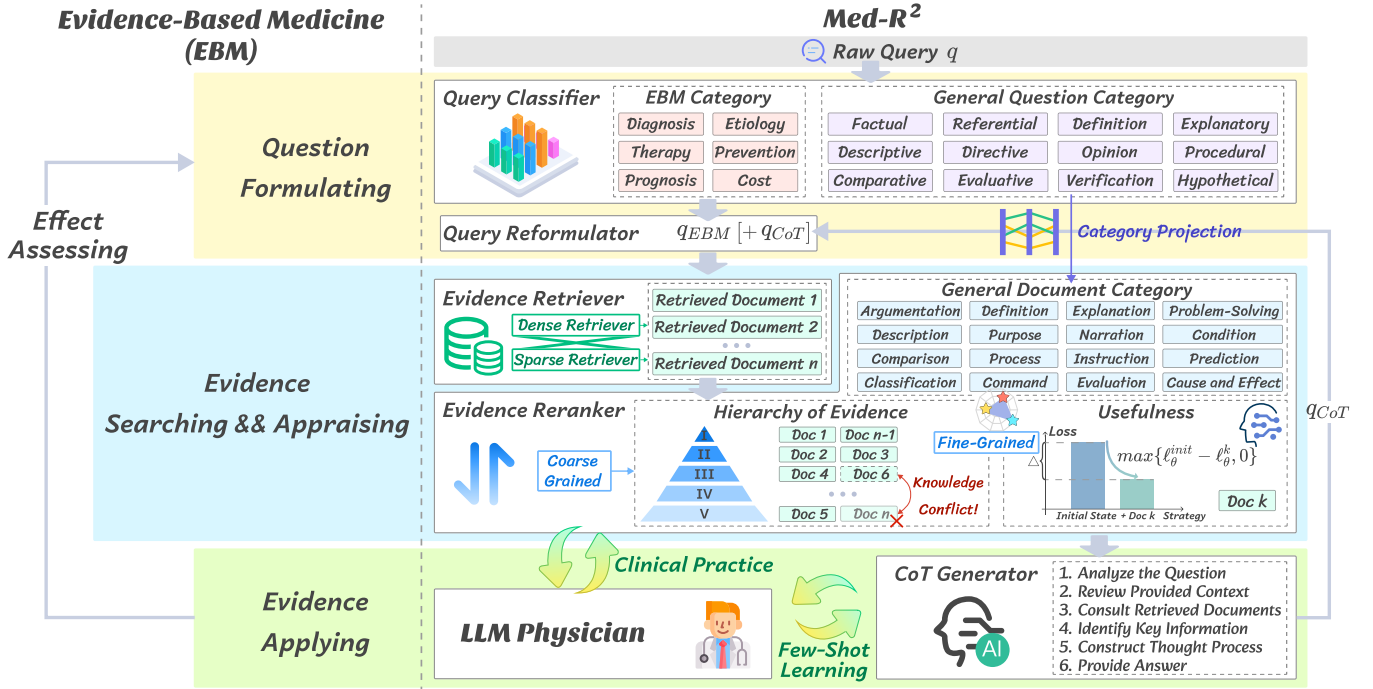


Figure 2: An illustration of Med-R²'s process, adhering to the Evidence-Based Medicine (EBM) workflow. We first categorize the query by EBM and general question types. Queries are then reformulated according to established EBM classification templates to ensure precision and relevance. In the evidence searching and appraising stages, we employ a coarse-to-fine strategy to retrieve, filter, and re-rank the evidence documents within the knowledge base. CoT sequences are then generated from processed evidence to refine retrieval space, iterating to ensure the robustness.

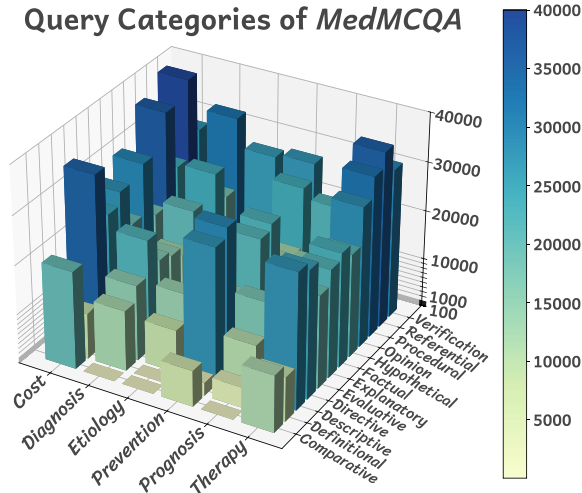


Figure 3: Query category of MedMCQA. We employ a logarithmic scale (base 10) on the z-axis, ranging from 1 to 40000.

general question categories, as illustrated in Figure 2. The application of this classification scheme to the MedMCQA dataset yielded the categorized results presented in Figure 3. For the classification task, we have manually annotated 100 samples to fine-tune

Qwen2.5-72B-Instruct¹, and employed the trained model as our classifier. We perform domain-specific reformulations of the original queries based on their respective classes of the EBM categories to align with the professional context. Meanwhile, the general question categories are utilized as one of the criteria to rerank retrieved documents, thereby prioritizing those that best match the current query's intent and document type preferences described in Figure 5.

3.2 Evidence Searching & Appraising

The retrieval and appraisal of evidence constitute one of the most critical stages in EBM. The professional reformulating of queries in the preceding stages (Section 3.1) aims to enhance the precision of retrieving relevant evidence documents. This stage comprises two key components: **evidence retriever** and **evidence reranker**.

3.2.1 Evidence Retriever. To construct a more comprehensive medical knowledge base adaptable to diverse healthcare scenarios, we have amassed a collection of medical data to build our **knowledge corpus**. The final medical knowledge base employed for retrieval comprises four distinct types of resources: *academic papers*, *entries*, *books*, and *guidelines*, with details of the data sources and statistics depicted in Table 1. For resources with extensive content such as academic papers, books, and guidelines, we first perform content segmentation with a threshold set at 10,000 tokens. We prioritize

¹<https://huggingface.co/Qwen/Qwen2.5-72B-Instruct>

Table 1: Overall statistics of medical knowledge resources.

Source Type	#Volume	#Tokens / Doc			
		Max	Min	Mean	Medium
Academic Papers	600,000	10,643	279	3,820	3,097
Entries	470,000	6,538	56	1,387	1,962
Books	10,000	15,384	524	4,083	4,319
Guidelines	10,000	4,981	74	1,100	1,778
Total	1,090,000	15,384	56	2,748	2,219

dividing the content based on natural chapters. If natural chapters cannot be identified or exceed the threshold, we resort to truncation according to the predefined limit.

We integrate multiple types of retrievers to optimize our retrieval performance, including BGE-Large-EN-v1.5² for dense retrieval and SPLADE-v3³ for sparse retrieval. We then consolidate the documents retrieved through both methods into a unified collection that contains n documents, $\mathcal{D} = \{d_i\}_{i=1}^n = \mathcal{D}^D \cup \mathcal{D}^S$, where $\mathcal{D}^D$ comprises documents obtained by dense retrieval, and $\mathcal{D}^S$ represents those acquired via sparse retrieval.

3.2.2 Evidence Reranker. For the document collections retrieved in Section 3.2.1, we employ a *coarse-to-fine* strategy for reranking. We initially utilize the BGE-Reranker-v2-M3⁴ to rerank $\mathcal{D} = \{d_i\}_{i=1}^n$ based on their semantic relevance to the current query at a *coarse* granularity, returning the top k documents⁵, denoted as $\mathcal{S}^c = \{d_i\}_{i=1}^k$. At this stage, the total length of these k documents significantly exceeds the context limit of the model. Then we conduct a *fine*-grained reranking that integrates three distinct criteria for the current k documents to get $\mathcal{S}^f = \{d_i\}_{i=1}^k$, enabling the model to provide answers based on the most effective retrieved documents within the limited length of the context window. The fine-grained reranking score can be formulated as:

$$\mathcal{F}(x) = f_h(x) \cdot f_g(x) (1 + \alpha \cdot f_u(x)) \quad (1)$$

where $f_h(x)$, $f_u(x)$ and $f_g(x)$ refer to scores of each refined evaluation criteria, while α is the non-negative hyper-parameter for weight controlling. The detailed implementation and explanation for the derivation of Equation (1) can be found in Section B.3.

Hierarchy of Evidence Medical knowledge encompasses facts and theories that are not always consistent, and sometimes even contradictory. This criterion serves dual purposes: *scoring* and *conflict filtering*. The recalled documents that have undergone coarse-grained reranking are first categorized according to their evidence levels. Subsequently, contents of these documents are analyzed for conflicting facts, where the documents that contain conflicting facts and have lower evidence ratings are filtered out. Formally, each retrieved document $d_i \in \mathcal{S}^c$ is associated with an integer evidence level e , where $e \in \{x \in \mathbb{Z} \mid 1 \leq x \leq 9\}$, with 1 indicating the highest level of credibility, as depicted in Figure 4:

$$f_h(x) = 9 - (e_x - 1) \quad (2)$$

²<https://huggingface.co/BAAI/bge-large-en-v1.5>

³<https://huggingface.co/naver/splade-v3>

⁴<https://huggingface.co/BAAI/bge-reranker-v2-m3>

⁵During the previous retrieval phase, documents of all retrieval types are subjected to a *unified ranking*, within which the top- k documents are selected for further processing.

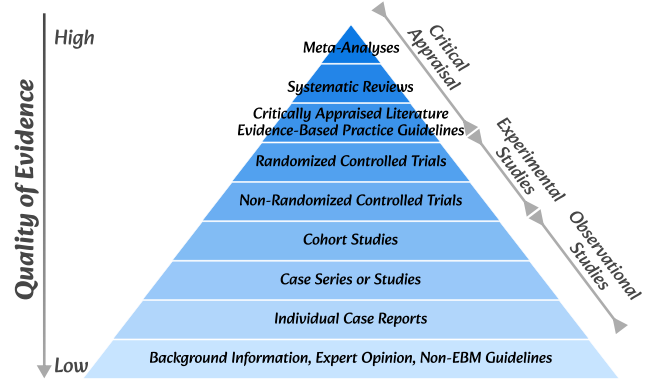


Figure 4: Hierarchy of evidence. The base of the pyramid represents the lowest quality, while the apex denotes the highest.

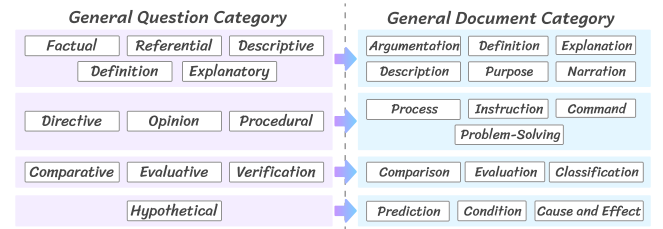


Figure 5: Query Document Projection.

By applying the evidence assessment criteria, the retrieved documents are categorized into multiple evidence levels, and professional analysis of evidence at different levels of authority is performed, preventing misjudgments caused by information clutter.

Usefulness The usefulness ranker is employed to assess the contribution of retrieved documents to the answering process. Specifically, we quantify the usefulness of a document by measuring the difference in loss before and after using the retrieved document to answer the question. This is achieved with a lightweight proxy model that evaluates the impact of the document on the answer’s quality, which can be written as:

$$f_u(x) = \max \{ \ell_{\theta}^{init} - \ell_{\theta}^x, 0 \} \quad (3)$$

where ℓ_{θ}^{init} indicates the loss without referring to any retrieved documents, while ℓ_{θ}^x represents that informed by document x .

General Document Category It corresponds to the mapping of *general natural language question* types discussed in Section 3.1. Different categories of questions desire distinct answer structures. For instance, questions regarding procedural steps are ideally answered by documents that describe processes rather than those that define concepts. To address this, we categorize the retrieved documents into 16 document types, denoted as C , as outlined in Table 6, and score them based on their alignment with the response type preferred by the original query, as depicted in Figure 5. The scoring function is defined as:

$$f_g(x) = \sum_{j=1}^{|C^e|} p(x|c_j) \quad (4)$$

Algorithm 1 Scoring Based on General Document Category Classification

Input: Coarsely reranked documents $\mathcal{S}^c$, Query-document mapping $\mathcal{A}$ based on Figure 5, classifier M_P , List of general document categories C
Parameter: List of current expected document categories C^e , $C^e \subset C$
Output: Scores based on document category $f_g(S^c)$
Define q^c : category of the query used for retrieving
Define $\tilde{p}$: category probability distribution of document d

- 1: **for** each document d_i in S^c **do**
- 2: /* Step 1: Expected Category Mapping */
- 3: Obtain expected categories $C^e = \mathcal{A}(q_{d_i}^c)$, where $C^e \subset C$
- 4: /* Step 2: Document Category Probability Inference */
- 5: Provide category probability distribution of d_i referring to M_P :

$$\tilde{p}_i = \{p(d_i|c_j)\}_{j=1}^{|C|} \leftarrow M_P(d_i)$$

- 6: /* Step 3: Statistics Aggregation */
- 7: Calculate the sum of probabilities across all expected categories:

$$f_g(d_i) = \sum_{j=1}^{|C^e|} p(d_i|c_j), \text{ where } c_j \in C^e$$
- 8: **end for**
- 9: **Return** $f_g(S^c) = \{f_g(d_i)\}_{i=1}^{|S^c|}$

where $C^e = \{c_j\}_{j=1}^{|C^e|}$ stands for the list of expected document types ($C^e \subset C$), and $p(x|c_j)$ represents the probability that the document x belongs to the current expected type c_j from C^e . Details of computational procedures are provided in Algorithm 1.

3.3 Evidence Applying

Through the comprehensive evaluation and reranking of retrieved documentary evidence in Section 3.2.2, we aim to ensure the application of the highest-quality available evidence to decision-making in the medical field. This process extends beyond merely applying research findings, where integrating the professional reasoning and judgment is essential. Therefore, it is imperative to make *professional and reasoned inferences* based on the retrieved documents.

CoT Generator This module constructs a chain-of-thought reasoning process based on the original medical query and the retrieved evidence documents⁶. It serves dual functions: (1) *Component of Query Reformulation*: It contributes to the subsequent query reformulation process, facilitating the retrieval of evidence documents relevant to the question. In Section 5, we also clarify the rationale behind not incorporating models' generated CoT sequences into the query reformulator for medical knowledge retrieval at the initial iteration. (2) *Few-Shot Learning Instance*: It provides few-shot examples for the LLM physician (target model) intended for downstream task evaluating, demonstrating how to analyze the retrieved evidence, identify the key information, and address medical queries effectively.

3.4 Effect Assessment

Our assessment of evidence is centered on the following aspect: the *stability* of evidence document retrieval across different stages. Algorithm 2 delineates the evidence evaluation process and the

⁶When ablating the integration of CoT sequences at the onset in Section 5, the CoT sequences are curated solely from the original medical query during *the initial iteration*, excluding the retrieved evidence documents.

Algorithm 2 Evidence Assessment and Iterative Retrieval Loop for Query Corpus

Input: Query corpus Q , CoT generator M_{CoT} , Context window length w , Hyperparameters: alteration threshold δ , maximum iterations T
Params: Coarsely reranked documents $\mathcal{S}^c$, finely reranked documents $\mathcal{S}^f$
Output: Selected evidence documents $\mathcal{S}_{Q-top}^f$ and CoT sequences Q^{CoT}
Define q^{CoT} : chain-of-thought sequence generated based on query and associated evidence documents
Define $E(d)$: embedding of document d

- 1: $\mathcal{S}_{Q-top}^f, Q^{CoT} \leftarrow [], []$
- 2: **for** each query q_i in Q **do**
- 3: **for** $t = 1, 2, \dots, T$ **do**
- 4: Reformulate query q_i (Section 3.1), search and appraise the retrieved evidence (Section 3.2)
- 5: /* Step 1: CoT Generation */
- 6: Select top k documents from $\mathcal{S}_i^{f(t)}$ referring to w :

$$\mathcal{S}_{i-top}^{f(t)} \leftarrow \text{SelectTopK}(\mathcal{S}_i^{f(t)}, k, w)$$
- 7: Generate chain-of-thought sequence based on q_i and $\mathcal{S}_{i-top}^{f(t)}$:

$$q_i^{CoT(t)} \leftarrow M_{CoT}(q_i, \mathcal{S}_{i-top}^{f(t)})$$
- 8: /* Step 2: Evidence Assessment */
- 9: Compute semantic stability of docs in consecutive iterations:

$$\tilde{\mu}_i^{(t)} \leftarrow \left\| \frac{1}{|\mathcal{S}_i^{f(t)}|} \sum_{d_i \in \mathcal{S}_i^{f(t)}} E(d_i) - \frac{1}{|\mathcal{S}_i^{f(t-1)}|} \sum_{d_i \in \mathcal{S}_i^{f(t-1)}} E(d_i) \right\|$$
- 10: /* Step 3: Termination Condition Evaluation */
- 11: **if** $\tilde{\mu}_i^{(t)} < \delta$ **then**
- 12: $\mathcal{S}_{Q-top}^f.append(\mathcal{S}_{i-top}^{f(t)}), Q^{CoT}.append(q_i^{CoT(t)})$
- 13: **Break**
- 14: **end if**
- 15: **end for**
- 16: $\mathcal{S}_{Q-top}^f.append(\mathcal{S}_{i-top}^{f(T)}), Q^{CoT}.append(q_i^{CoT(T)})$
- 17: **end for**
- 18: **Return** Selected documents $\mathcal{S}_{Q-top}^f$ and CoT sequences Q^{CoT}

iterative retrieval loop for the query corpus within the pipeline, where we determine the number of iterations for optimization termination based on our assessment of such factor. This ensures that when evaluating the LLM physician (target model) on medical tasks, the retrieved documents provided are effective and robust.

4 Experiments and Results

4.1 Experimental Setup

Model Details. We employ the open-sourced LLMs from LLaMA [8, 40] and Qwen [47] series as our target models for evaluation. We assessed models including LLaMA3.1-8B, Qwen2.5-14B, Qwen2.5-32B and LLaMA3.1-70B, scaling from 8B to 70B. The default setting of context window for our main experiments is 4K, with an in-depth scaling analysis presented in Section 5.

Datasets. We have selected eight medical datasets including PubMedQA [21], MedQA-USMLE, MedQA-MCML [20], MedM-CQA [33], MMLU-Med [15], NEJM [25], MedXpertQA [52], and

Table 2: Comparison of Med-R² with baselines. The best and second best are in bold and underlined.

Model	Method	MedQA-USMLE	MedQA-MCMLE	MedMCQA	PubMedQA	MMLU-Med	NEJMQA	MedXpertQA	RareArena-RDC	RareArena-RDS	Avg.
		Within-Dataset Fine-Tuning				Cross-Dataset Fine-Tuning					
Frontier Models											
GPT-4o Claude3.5-Sonnet DeepSeek-V3	–	84.95	79.28	74.60	77.20	84.44	72.20	30.37	73.76	47.50	69.31
	–	83.20	72.54	68.80	76.40	83.90	68.52	21.31	68.46	50.78	65.99
	–	80.93	77.82	74.30	73.60	87.67	74.23	24.16	75.09	45.72	68.73
Open-Sourced Medical Models											
PMC-LLaMA-7B	–	38.65	35.30	32.40	52.26	26.87	35.63	11.02	28.96	21.67	31.42
MEDITRON-7B	–	46.76	42.78	36.83	58.32	43.60	38.80	11.74	35.68	26.50	37.89
PMC-LLaMA-13B	–	40.38	33.07	37.55	54.63	46.85	40.60	11.44	39.86	28.98	37.04
MEDITRON-70B	–	55.05	52.21	53.50	72.84	71.45	66.97	18.05	74.81	46.58	56.83
Open-Sourced Base Models											
LLaMA3.1-8B	Direct Response	31.16	41.45	30.02	36.17	37.12	50.41	14.90	41.85	22.16	33.91
	Vanilla RAG	55.80	59.38	35.91	47.10	43.54	52.01	15.50	56.23	35.06	44.51
	Fine-Tuning	76.53	85.24	47.38	42.78	42.92	52.48	15.54	55.38	38.37	50.73
	LLM-AMT	52.91	66.63	45.08	44.60	45.39	53.07	16.22	60.20	40.29	47.16
	Med-R ² (ours)	77.01	84.16	52.33	52.75	45.19	55.69	17.79	62.90	44.82	54.74
Qwen2.5-14B	Direct Response	50.01	65.23	42.85	56.93	71.60	45.63	11.06	43.17	26.58	45.89
	Vanilla RAG	54.88	75.60	42.06	60.38	79.46	45.69	12.28	54.00	28.05	50.27
	Fine-Tuning	55.87	85.52	47.64	54.29	80.41	46.67	12.30	66.32	43.66	54.75
	LLM-AMT	51.48	77.11	43.09	62.42	78.78	47.44	13.73	60.03	38.81	52.54
	Med-R ² (ours)	54.34	80.19	48.36	68.32	84.03	48.03	14.45	72.80	46.30	57.43
Qwen2.5-32B	Direct Response	16.23	87.07	66.44	68.66	80.19	43.59	15.06	59.67	31.33	52.03
	Vanilla RAG	19.33	89.30	67.63	67.06	83.85	46.63	17.49	66.10	40.59	55.33
	Fine-Tuning	25.57	89.97	66.17	66.69	82.40	52.58	15.71	70.20	45.08	57.15
	LLM-AMT	19.38	88.08	68.33	68.61	82.06	46.82	17.90	66.75	45.40	55.93
	Med-R ² (ours)	24.43	90.01	69.09	68.36	84.95	50.40	18.54	72.51	55.36	59.29
LLaMA3.1-70B	Direct Response	46.43	58.36	62.33	66.81	71.33	65.43	23.43	72.59	43.54	56.69
	Vanilla RAG	62.66	77.91	66.63	68.78	76.95	66.40	25.95	77.70	44.88	63.10
	Fine-Tuning	87.17	86.21	71.65	74.72	79.46	68.11	24.18	76.41	47.23	68.35
	LLM-AMT	79.18	68.59	70.12	79.05	80.74	68.90	26.53	77.00	45.87	66.22
	Med-R ² (ours)	86.37	84.58	73.36	78.24	82.82	70.18	26.84	78.60	49.42	70.04

RareArena [39], covering both standard and real-world clinical scenarios. We use accuracy as evaluation metrics.

Implementation. We constructed the medical knowledge corpus by establishing FAISS vector library [23]. Experiments related to model training were conducted based on full-parameter fine-tuning, during which we utilized a learning rate scheduler featuring linear warm-up and cosine decay, peaking at a learning rate of 2e-5, alongside a warmup ratio of 0.03, a weight decay of 0.0 and a batch size of 128 for 3 epochs. We conducted all training and evaluation experiments on NVIDIA RTX H800 GPUs with 80G memory.

4.2 Baselines

We compare Med-R² with the following baselines: (1) **Frontier Models** contain GPT-4o [16], Claude3.5-Sonnet [2] and DeepSeek-V3 [29]. (2) **Open-Sourced Medical Models** include PMC-LLaMA-7B, PMC-LLaMA-13B [43], MEDITRON-7B and MEDITRON-70B [6]. (3) The simplest baseline is **Direct Response**, where the model answer medical questions directly without the aid of external knowledge bases or dataset fine-tuning. (4) **Vanilla RAG** [27] utilizes raw queries for evidence searching, and the retrieved documents are then directly integrated into the generation process without any further manipulation. (5) **Fine-Tuning** leverages medical datasets to further train the model under supervised conditions. Here we employ two strategies: *within-dataset fine-tuning*, where the datasets for training and evaluation are derived from different parts of the

same data corpus, and *cross-dataset fine-tuning*, where the model is fine-tuned on one medical dataset (e.g., MedMCQA) and then evaluated on different datasets (e.g., PubMedQA). (6) **LLM-AMT** [42] is a dedicated process tailored for biomedical question answering, which includes typical modules such as query augementer, hybrid retriever, knowledge refiner, etc. Details are discussed in Section C.2.

4.3 Main Results

Med-R² is effective across different models. Table 2 shows that the incorporation of external knowledge bases significantly enhances the model’s ability to address medical queries, where even *vanilla RAG* depicts an average enhancement of 13.10% over *direct responses*. Furthermore, Med-R² provides an added layer by adhering to the EBM process, achieving an average improvement of 28.10% over the *direct response* strategy. Notably, for lightweight models such as LLaMA3.1-8B, Med-R² demonstrates increases of 61.43%. We surmise that while lightweight models inherently lack comprehensive domain-specific medical knowledge, they possess the capability to read and identify information from external medical documents. Consequently, effective augmentation from external knowledge substantially bolsters the models’ capacity to tackle medical-domain questions. Compared with medical-specific models of the same parameter scale, models equipped with Med-R² demonstrate superior performance in medical tasks. Moreover, LLaMA3.1-70B + Med-R² has the potential to surpass frontier models under

medical scenarios, achieving average improvements over GPT-4o, Claude3.5-Sonnet and DeepSeek-V3 by 1.05%, 6.14% and 1.91%.

Med-R² shows superiority compared to fine-tuning. From Table 2, we find that Med-R² stands out as the only approach among those leveraging external knowledge that surpasses the average performance of *fine-tuning* methods. Specifically, Med-R² exhibits nearly equivalent performance to fine-tuning strategies in *within-dataset training*, yet it significantly outperforms in *cross-dataset training*, achieving an enhancement of 7.61% and an overall capability improvement of 4.55%. One contributing factor is that during within-dataset fine-tuning, the training and testing datasets are of the same origin, thus a model trained on homogeneous data could achieve substantial performance gains on the test set. Conversely, in cross-dataset fine-tuning, the heterogeneity between the training and testing datasets more rigorously assesses the model’s ability of generalization. Under these circumstances, the utilization of a comprehensive external medical knowledge base and the effective retrieval and extraction of pertinent information becomes particularly crucial. Additionally, considering that fine-tuning necessitates additional training time and computational resources, Med-R² emerges as a more efficient approach.

5 Ablations and Analysis Across Scales

We further analyze the impact of *model scale* and *context window length* on Med-R², and provide rationale for incorporating the CoT sequences of models into the retrieval process starting from the second iteration. Additionally, we have also ablate the components of Med-R² in Table 3 to identify the contribution of each module.

Integrating CoT at the onset may bring adverse effects. We ablate the components of query reformulator $q_{EBM} [+q_{CoT}]$ through integrating the chain-of-thought (CoT) sequences generated by models into the initial evidence retrieval phase, which is denoted as **Med-R²-CoT**. Figure 6 presents the medical task performance of Med-R²-CoT and all baselines included in our main experiments. It is observed that Med-R²-CoT exhibits a decline in performance compared to Med-R², with the disparity increasing as the model parameter scale decreases. Notably, at the 8 billion parameter level (e.g., LLaMA3.1-8B), the performance of Med-R²-CoT is even inferior to that of the *vanilla* RAG strategy. We hypothesize that one contributing factor is that the lightweight models’ less solid grasp of medical knowledge. As a result, in the absence of external medical knowledge, models of these scales may struggle to align the direction of their thinking with the original query, and thus, the generated CoT sequences may even negatively impact the retrieval effectiveness. However, this phenomenon is somewhat mitigated as the model parameter scale increases.

Effect of context window scale on model’s performance. We compared the performance of Med-R² across models with varying parameter scales and different context window lengths on medical tasks, and then plotted heatmaps illustrating the percentage improvement of Med-R² over *direct responses*, as depicted in Figure 7. It reveals that Med-R² exhibits an optimal context window length for enhancing the model’s medical performance, which increases with the growth of model parameter size. Concurrently, the enhancement of Med-R² follows a trend of initial decline followed by an increase with the escalation of model scale. Specifically:

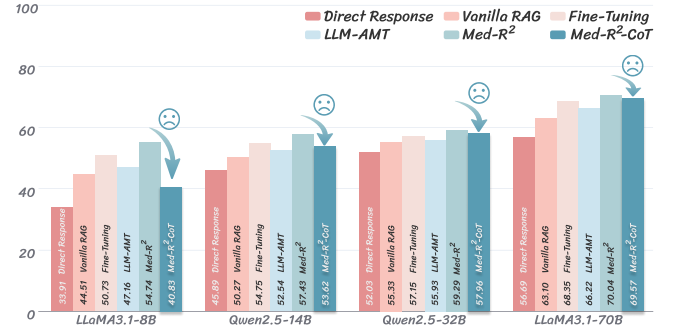


Figure 6: Average evaluation results of baselines from Table 2 and Med-R²-CoT. Compared to Med-R², Med-R²-CoT involves the immediate incorporation of models’ CoT sequence into the query reformulator for retrieval during the initial round.

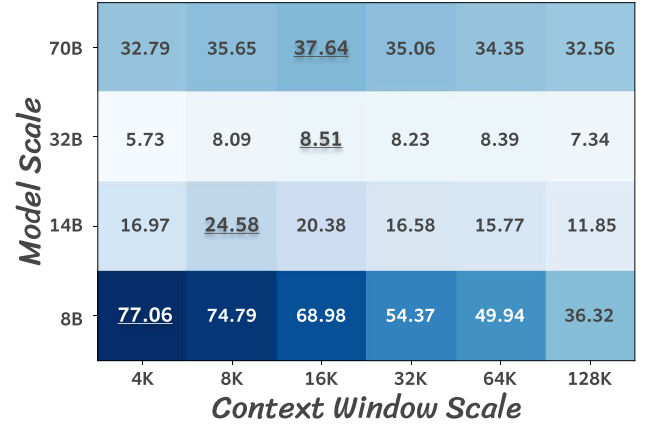


Figure 7: The percentage increase of performance achieved by Med-R² over direct responses across various context window sizes and model scales.

- For models of 8B parameters, the most pronounced enhancement was observed at a 4K context window, but this benefit diminished sharply as the context length increased.
- For 14B models, the 8K length stands out, where a measurable decrease in performances is observed as the context window expanded. Moreover, the improvement provided by Med-R² at this scale is *the most modest* compared to others.
- Models with 32B and 70B parameters achieved optimal performance at a 16K context length, demonstrating relatively stable improvement across various context window lengths.

We hypothesize that as the context length increases, the role of Med-R²’s reranker diminishes since most retrieved documents are fed into the same context window of the model. At this point, lightweight models, particularly those around 8B, which may not have a solid grasp of medical knowledge, and overly long sequences could reduce the model’s efficiency in extracting key information, potentially leading to the generation of hallucinations and adversely affecting the model’s medical performance. However, the addition of other modules, such as the query reformulator, can help improve the precision of knowledge retrieval, thereby mitigating this negative

Table 3: Module analysis of Med-R² . We sequentially integrate various modules onto the *vanilla* RAG systems to conduct comparative analyses. We highlight the optimal performance values for models with **single and **dual** component additions.**

Model	Method	MedQA-USMLE	MedQA-MCML	MedMCQA	PubMedQA	MMLU-Med	Average
LLaMA3.1-8B	Direct Response	31.16	41.45	30.02	36.17	37.12	35.18
	Vanilla RAG	55.80	59.38	35.91	47.10	43.54	48.35
	+ Query Reformulator	62.47	72.85	41.66	48.72	43.98	53.94
	+ Evidence Reranker	68.62	75.74	43.84	49.97	43.76	56.34
	+ CoT Generator	59.87	65.63	39.45	47.71	43.61	51.25
	+ Query Reformulator, Evidence Reranker	74.41	81.68	47.71	53.96	44.23	60.40
	+ Query Reformulator, CoT Generator	70.86	77.43	44.83	51.75	44.02	57.78
	+ Evidence Reranker, CoT Generator	72.69	79.84	45.12	52.25	44.00	58.78
	Med-R² (ours)	77.01	84.16	52.33	52.75	45.19	62.29
Qwen2.5-14B	Direct Response	50.01	65.23	42.85	56.93	71.60	57.32
	Vanilla RAG	54.88	75.60	42.06	60.38	79.46	62.48
	+ Query Reformulator	54.93	78.96	45.43	64.85	81.73	65.18
	+ Evidence Reranker	55.03	78.83	45.97	65.08	81.56	65.29
	+ CoT Generator	54.62	76.72	43.86	62.13	80.02	63.47
	+ Query Reformulator, Evidence Reranker	55.38	82.04	48.29	67.02	83.17	67.18
	+ Query Reformulator, CoT Generator	54.77	80.58	47.61	66.98	82.21	66.43
	+ Evidence Reranker, CoT Generator	54.81	81.79	47.87	67.48	82.95	66.98
	Med-R² (ours)	54.34	80.19	48.36	68.32	84.03	67.05
Qwen2.5-32B	Direct Response	16.23	87.07	66.44	68.66	80.19	63.72
	Vanilla RAG	19.33	89.30	67.63	67.06	83.85	65.43
	+ Query Reformulator	20.14	89.42	68.72	68.41	83.97	66.13
	+ Evidence Reranker	20.83	89.88	68.88	68.47	83.99	66.41
	+ CoT Generator	19.78	88.69	67.94	67.72	83.92	65.61
	+ Query Reformulator, Evidence Reranker	23.04	89.25	70.35	68.89	84.46	67.20
	+ Query Reformulator, CoT Generator	21.97	89.56	69.06	68.56	84.08	66.65
	+ Evidence Reranker, CoT Generator	23.01	89.75	69.75	69.00	84.39	67.18
	Med-R² (ours)	24.43	90.01	69.09	68.36	84.95	67.37
LLaMA3.1-70B	Direct Response	46.43	58.36	62.33	66.81	71.33	61.05
	Vanilla RAG	62.66	77.91	66.63	68.78	76.95	70.59
	+ Query Reformulator	70.82	80.64	69.72	71.82	73.56	73.31
	+ Evidence Reranker	73.96	80.78	70.87	74.64	75.28	75.11
	+ CoT Generator	68.08	78.65	68.85	70.09	73.72	71.88
	+ Query Reformulator, Evidence Reranker	80.41	84.09	73.01	77.80	80.68	79.20
	+ Query Reformulator, CoT Generator	78.62	82.56	72.71	76.06	78.56	77.70
	+ Evidence Reranker, CoT Generator	81.93	82.97	72.88	76.52	81.62	79.18
	Med-R² (ours)	86.37	84.58	73.36	78.24	82.82	81.07

impact to some extent. In contrast, models with 32B and higher scales exhibit greater robustness, and the impact of increasing the window length on Med-R²'s effectiveness is less pronounced.

Module Impact Analysis. We have analyzed the contributions of various modules in Med-R² to the model's performance in the medical domain using a default context window size of 4K, sequentially incorporating components into the vanilla RAG framework. As shown in Table 3, we decompose the modules into three components: **query reformulator**, **evidence reranker**, and **CoT generator**. Overall, the evidence reranker contributed the most to models' performances among the individual components. When combined with the query reformulator, the performance gains were even more pronounced, showing a synergistic effect. The addition of the CoT generator further enhanced the model's ability to effectively utilize retrieved medical documents, providing substantial added value.

6 Discussion

In this study, we follow the Evidence-Based Medicine (EBM) process to design a novel LLM physician framework, which effectively leverages the retrieval, filtering, and reasoning processes inherent to EBM. Experiments demonstrate that Med-R² holds superior advantages over existing strategies, while also reducing the substantial computational costs associated with model training. Furthermore, by scaling the model parameters and adjusting the context window length, we demonstrate the robustness of Med-R² in achieving optimal performance across a broad spectrum of medical tasks.

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Appendix

A Evidence-Based Medicine (EBM)

Evidence-Based Medicine (EBM) is defined as the conscientious, explicit, and judicious application of the best current evidence in making decisions regarding the care of individual patients. This practice entails the integration of individual clinical expertise with the most reliable external clinical evidence derived from systematic research [35]. The EBM process typically includes five stages, *question formulating, evidence searching, evidence appraising, evidence applying and effect assessing*, which aims to make the best possible health care decision through iterative improvements. The *hierarchy of evidence* guides the clinical decision-making, since not all evidence is created equal, as described in Figure 4.

B Method Details

B.1 Question Formulation Details

B.1.1 Details of Query Classification. We categorize the medical queries along two orthogonal dimensions: *Evidence-Based Medicine (EBM) categories* and *general natural language question types*.

Evidence-Based Medicine (EBM) Categories Nonprofessional queries may fail to clearly articulate the current medical symptoms, thereby hindering the effective retrieval of the necessary evidence. We classify medical queries into several types according to the categories of Evidence-Based Medicine (EBM) questions, including *diagnosis, therapy, prognosis, etiology, prevention, cost*, etc., and then conduct professional medical query reformulations based on their EBM categorization to emphasize specialized retrieval. Instructions for targeted augmentation of clinical queries are outlined in Table 5, where we employ Qwen2.5-72B-Instruct⁷ fine-tuned on our 100 manually annotated samples as our question reformulator.

General Natural Language Question Types It serves as a crucial reference for filtering and reranking the documents retrieved

⁷<https://huggingface.co/Qwen/Qwen2.5-72B-Instruct>

Table 4: Explanations and descriptions for each category within the general question classification.

Category	Details of Description
Factual	Inquiring into specific and objective facts or data.
Referential	Seeking answers by referencing specific documents, resources, or other information.
Definition	Inquiring about the definition or explanation of a concept or entity.
Explanatory	Seeking explanations for the causes of phenomena, processes, or events.
Descriptive	Requesting a description of the characteristics, properties, and features of an entity.
Directive	Seeking guidance or recommendations.
Opinion	Pertaining to individual feelings, attitudes, or preferences.
Procedural	Inquiring about the specific steps to complete a particular task or activity.
Comparative	Inquiring comparison of the differences between two or more entities.
Evaluative	Assessing the validity or quality of a statement or viewpoint.
Verification	Confirming or verifying the authenticity or accuracy of certain information.
Hypothetical	Presenting a hypothetical scenario and requesting predictions of outcomes.

Table 5: Prompts for query reformulation of each category within the Evidence-Based Medicine (EBM) categories.

Category	Instructions for Query Reformulation
Diagnosis	Specify the condition you need to diagnose and ask about the accuracy, sensitivity, or specificity of specific diagnostic tests.
Therapy	Specify the disease or symptom along with the therapy being considered, and inquire about its effectiveness, safety, or comparison with other therapies.
Prognosis	Specify the disease or condition and ask about long-term outcomes such as survival rates, recovery chances, or disease progression.
Etiology	Describe the health issue and ask about potential causes, including risk factors, pathogens, or genetic background.
Prevention	Specify the disease or health issue and ask about the effectiveness of preventive measures or recommendations.
Cost	Specify the medical intervention or service and ask about cost-effectiveness analyses, including direct and indirect costs and cost-effectiveness ratios.

in response to a query, since the emphasis of the expected answer varies with different question types. We categorize queries into 12 natural language classes, as illustrated in Table 4, establishing a mapping between the types of questions and that of evidence to form a question-answer typology (depicted in Figure 5).

B.1.2 Details of Query Reformulator. Following the instructions outlined in Table 5, we have manually annotated 120 samples (20 samples for each EBM category) to fine-tune the Qwen2.5-72B-Instruct model, and employed the trained model to professionally reformulate the queries within the medical datasets to enhance the subsequent retrieval of relevant evidence pertaining to the current query from the knowledge corpus.

B.2 Knowledge Corpus Details

We organized the medical knowledge corpus from four data sources: *academic papers, entries, books, and guidelines*, as listed below:

- **Academic Papers** Academic literature provide valuable insights derived from the latest scientific investigations, offering a robust theoretical foundation for guiding clinical practice and informing public health decisions. We obtained the publicly available literature from PubMed Central (PMC)⁸, and extracted valid contents following the processing pipeline of Hakala et al. [13].
- **Entries** Medical entries provide valuable information across multiple dimensions of healthcare, from direct patient care to cutting-edge research. We extracted and curated medical-related data from the Wikipedia dataset⁹ to build the entries. In the end, we got approximately 470k documents.
- **Books** Medical textbooks are vital resources for medical knowledge retrieval, which can be consulted when faced with complex

⁸<https://pmc.ncbi.nlm.nih.gov/>

⁹<https://huggingface.co/datasets/wikimedia/wikipedia>

cases or when seeking updated knowledge about specific conditions. We gathered books from NCBI Bookshelf¹⁰, and collected 10k documents following the process of Hakala et al. [13].

- **Guidelines** Clinical practice guidelines serve as essential tools in Evidence-Based Medicine (EBM), aiding healthcare providers in making informed decisions for diagnosis and treatment. We utilized the guideline data when training MEDITRON series from Chen et al. [6], and gained approximately 10k documents.

B.3 Evidence Reranking Details

We detail the scoring clarification and implementation for the fine-grained reranking phase of retrieved documents from Equation (1).

- **Hierarchy of Evidence $f_h(x)$** : It should be clarified that for the *guideline-sourced* data retrieved from the knowledge corpus, we assign the f_h score directly to 7 ($e = 3$) based on the evidence level depicted in Figure 4. It is because the grading system of evidence hierarchy explicitly designates a specific level for guideline data.
- **Usefulness $f_u(x)$** : **1)** Regarding the *lightweight proxy models* we employed for scoring usefulness f_u of the retrieved documents, for the LLaMA series, we used the Sheared-LLaMA-1.3B [44] as a lightweight reference model; for the Qwen series, we utilized Qwen-2.5-1.5B [47]. **2)** Moreover, to *avoid potential label leakage* when computing f_u in Equation (3), we have implemented strategies to ensure that the entire framework does not have access to ground truth answers (labels) during the deployment phase, where we employed frontier models' responses to serve as the "reference answers" for consultation throughout the framework's operation. Specifically, we utilized the training subsets from MedQA-USMLE, MedQA-MCMLE [20], and MedMCQA [33] to fine-tune the Qwen2.5-72B-Instruct¹¹ model, resulting in model M_F . We replaced the labels with responses generated by model M_F for the current medical question.
- **General Document Category $f_g(x)$** : The calculation is based on the categorization and mapping to general document categories, as depicted in Algorithm 1. For this categorization, advanced language models provide *probabilities*. In addition, the relationship between the general question category and the general document category is *multifaceted* (illustrated in Figure 5), which further enhances the robustness of category assignment.

B.4 Evidence Assessment Details

Here, we provide the details of the iterative multi-document retrieval process, as illustrated in Algorithm 2.

- **Integration of CoT Sequences** It is important to note that during the initial retrieval phase, we *did not* incorporate the generated CoT sequences as part of the query reformulation $q_{EBM} [+q_{CoT}]$. Instead, this component was integrated in the *second* iteration, where we have shown the rationale in Section 5.

C Experiments Details

C.1 Details of Medical Datasets

For MedQA-USMLE and MedQA-MCMLE, the original data is divided into three parts: train, dev, and test. We utilize the training

¹⁰<https://www.ncbi.nlm.nih.gov/books/>

¹¹<https://huggingface.co/Qwen/Qwen2.5-72B-Instruct>

Table 6: Explanations and descriptions for each category within the general document classification.

Category	Details of Description
Argumentation	Presenting a viewpoint or argument, potentially accompanied by supporting evidence.
Definition	Providing a clear definition of a term or concept.
Description	Describing the characteristics or attributes of an object or event.
Explanation	Explaining a concept, process, or cause.
Purpose	Elucidating the purpose or intent behind a particular action or event.
Narration	Providing a narrative account of an event, experience, or story.
Process	Describing a process or a sequence of steps.
Instruction	Providing steps or guidance for executing a task or operation.
Command	Conveying a request that requires the listener to take action.
Problem-Solving	Proposing methods or strategies for addressing specific issues.
Comparison	Comparing the similarities or differences between two or more entities.
Evaluation	Articulating a judgment on a particular subject or behavior.
Classification	Categorizing objects or concepts into specific categories systems.
Condition	Describing the assumptions under which a particular event occurs.
Prediction	Forecasting future events or trends.
Cause and Effect	Describing the causal relationships between events.

part directly for model fine-tuning. The dev and test subsets are merged for evaluation, from which we selected 10 (or 11) instances as Chain-of-Thought (CoT) demonstration examples. Since the test portion of MedMCQA does not provide ground truth answers, we use the development set for evaluation. We employ MedQA-USMLE, MedQA-MCMLE, and MedMCQA datasets for *within-dataset fine-tuning*. For instance, we train with the training subset of MedQA-USMLE and subsequently evaluate with its corresponding test partitions. We utilize the remaining datasets for *cross-dataset fine-tuning* setting to test the model's generalizability on disparate data sources.

Table 7: Overall statistics of datasets. We utilize MedQA-USMLE, MedQA-MCMLE and MedMCQA for *within-dataset fine-tuning*, while others for *cross-dataset fine-tuning*.

Dataset	# Training Instance	# Testing Instance	# N-Shot for CoT	# Total Instance
MedQA-USMLE [20]	10178	2535	10	12723
MedQA-MCMLE [20]	27400	6840	11	34251
MedMCQA [33]	182822	4170	13	187005
PubMedQA [21]	-	990	10	1000
MMLU-Med [15]	-	1080	9	1089
NEJMQA [25]	-	655	10	655
MedXpertQA [52]	-	2455	10	2455
RareArena [39]	-	72,661	10	72,661

C.2 Details of Baselines

- **Frontier Models**: Frontier models are considered to represent the pinnacle of performance across various dimensions of LLMs, serving as the strongest baselines. Here, we have selected several state-of-the-art LLMs to establish the maximum potential of model performance on several medical benchmarks, including GPT-4o [16], Claude3.5-Sonnet [2], and DeepSeek-V3 [29].
- **Open-Sourced Medical Models**: These models stand for the domain-specific models that were specifically trained on medical data. We have selected PMC-LLaMA-7B, PMC-LLaMA-13B [43], MEDITRON-7B and MEDITRON-70B [6] to represent the open-sourced medical models for comparison to assess Med-R²'s relative advantage compared to specialized medical AI systems.
- **Direct Response**: We employ the base model to directly respond to medical queries without aid of additional training on any dataset or augmentation from external knowledge bases.

- **Vanilla RAG** [27]: It represents the most traditional and fundamental strategy for utilizing external knowledge bases to assist models in answering questions. In this study, we employ the same medical knowledge base as Med-R², but directly utilizing the raw text retrieved by FAISS and combining it with the original query to test the model's performance on medical tasks.
- **Within-Dataset Fine-Tuning**: The training and test sets originate from the same distribution, and models exhibit strong performance gains due to the homogeneity of the datasets.
- **Cross-Dataset Fine-Tuning**: It poses a greater challenge by testing the model's ability to generalize across different distributions, demanding robust transfer learning capabilities, which are bolstered by the integration of rich external medical knowledge.
- **LLM-AMT** [42]: LLM-AMT is a RAG system specifically designed for clinical question answering. It incorporates common RAG components such as a query augmentor, textbook retriever, knowledge refiner, and an LLM reader. LLM-AMT leverages a collection of medical textbooks as an external indexable medical knowledge base. However, in this comparison, we aim to assess the RAG's capability to retrieve, filter, and apply evidence from the same external knowledge base. To control variables, we employ the LLM-AMT's process but replace the medical knowledge base with our own constructed medical retrieval corpus.

C.3 Hyper-Parameters Setting

In the fine-grained reranking phase, we treat each factor as having equal importance, hence we set the weight controlling hyperparameter α in Equation (1) to 1. For retrieval iteration settings, we performed iterative retrieval on each query within the datasets in Table 7. Our analysis revealed that after approximately 5 iterations, the distribution of the retrieved document vectors stabilizes. Consequently, we set the maximum iterations T to 5 in Algorithm 2. After performing iterative retrieval for all samples in the dataset and calculating the average of the minimum distances in the retrieval space during iterations, we obtained a value of 6.85, which we adopted as our termination alteration threshold δ in Algorithm 2.

D Prompts

We present the prompts employed throughout our pipeline in Med-R². Specially, prompts for query reformulation according to the Evidence-Based Medicine (EBM) categories can be found in Table 5.

Prompt: Evidence-Based Medicine (EBM) Category Classification

You are an expert in sentence annotation within the medical field. There are 6 categories of clinical questions: Prognosis, Therapy, Etiology, Diagnosis, Prevention, and Cost. Please classify the following text fragment based on their purpose and structure by providing only the category name without additional commentary:

Question
{question}

Prompt: General Natural Language Question Category Classification

You are an expert in natural language question annotation. Given the following 12 categories of question types: Factual, Definitional, Explanatory, Descriptive, Directive, Opinion, Comparative, Evaluative, Hypothetical, Procedural, Referential, and Verification. Please classify the following text fragment based on their purpose and structure by providing only the category name without additional commentary:

Question
{question}

Prompt: Retrieved Document Category Classification

You are an expert in sentence annotation within the medical field. There are 16 categories of documents: Argumentation, Definition, Description, Explanation, Purpose, Narration, Process, Instruction, Command, Problem-Solving, Comparison, Evaluation, Classification, Condition, Prediction, Cause-and-Effect. Please classify the following text fragment based on their purpose and structure by providing the probability distribution of its belonging to each category, where the sum of probabilities across all categories equals 1, without additional commentary:

Document
{retrieved_document}

Output Format:

```
```json
{
 "Argumentation": "",
 "Definition": "",
 ...
}
```

### Prompt: Hierarchy of Evidence Judgement

You are an expert in evidence quality annotation within the medical field. There are 9 quality levels of evidence, ranging from the highest to the lowest as follows: Meta-Analyses, Systematic Reviews, Evidence-Based Practice Guidelines, Randomized Controlled Trials, Non-Randomized Controlled Trials, Cohort Studies, Case Series or Studies, Individual Case Reports, Expert Opinion. Please classify the following evidence document based on its structure and characteristics, providing only the names of the levels, without any additional description:

# **Evidence**  
{retrieved\_document}